



ΠΟΛΥΤΕΧΝΕΙΟ ΚΡΗΤΗΣ
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Quantum Cryptography

Quantum Key Distribution

E91 Protocol

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1.1 What is cryptography

Cryptography is a separate field of science whose main focus is securing communication and ensuring privacy between parties. To understand what cryptography aims to achieve, imagine trying to have a private conversation with someone in a public space without wanting anyone else to understand you.

To make this possible, scientists have developed crypto algorithms based on mathematical principles that can ensure privacy between parties. These algorithms work by transforming a message in a way that only the receiver can recover to its original shape. The privacy that these algorithms provide depends on a single factor known as, the secrecy of the key. Each key or pair of keys, encrypts the plain text in a unique way, that is irreversible without the decryption key (can be the same, can be other based on the algorithm).

1.2 The need of Key distribution algorithms

The fundamental concept in modern cryptography is that we cannot rely on the secrecy of the algorithm itself, as they can often be reverse-engineered, that's why we follow the principle that the algorithm is known and the key is the only unknown, also known as Kerckhoffs's principle.

Therefore, it becomes essential to develop secure methods for key distribution mechanisms that allow parties to exchange keys safely, even over insecure channels. Channels such as the internet do not guarantee privacy between the communicating parties. It is very easy for an attacker, known as a man-in-the-middle, to intercept network traffic and potentially capture the encryption key, without even get noticed.

1.3 Why is cryptography so necessary

Almost all of our daily activities in the current digital age include electronic communication, from banking and online shopping to exchanging private information about individuals, organizations, or even the government, resulting in the requirement to provide security and privacy when transferring data electronically. A strong answer is provided by cryptography, which shields data from unwanted access.

Beyond securing the transmission of data, cryptographic algorithms also provide verification of the integrity and authenticity of the files. Cryptographic hash functions, which generate unique digital fingerprints for both the data and the sender, can verify the source and the integrity of the files, detecting unauthorized modifications to the data and ensuring the communication is performed with the right person.

The risk of data leak, identity theft, and cyberattacks grows significantly. Without effective cryptographic systems, private information could easily be intercepted, altered, or misused by third parties.

1.4 Notable cryptography algorithms

This section will cover the following essential categories of cryptographic algorithms which are necessary in order to accomplish the objectives mentioned above :

- Asymmetric and symmetric encryption, which are used to transform plaintext into unreadable cipher text to protect the data during transmission .
- Key distribution algorithms, which enable the secure exchange of cryptographic keys between communicating parties .
- Hash functions, which ensure data integrity by detecting any unauthorized alterations to the data.

Encryption algorithms :

We divide encryption algorithms in two sub categories asymmetric and symmetric . Asymmetric encryption is an encryption method that uses two different keys—a public key and a private key—to encrypt and decrypt data. It's generally regarded as more secure, though less efficient, than symmetric encryption. It works by creating a pair of keys, one public and one private. Anyone can use a public key to encrypt data. However, only the holders of the corresponding private key can decrypt that data. Symmetric encryption differs from asymmetric encryption because it uses a single key to encrypt and decrypt data .

Common asymmetric encryption algorithms :

- **Rivest-Shamir-Adleman (RSA)**
RSA is an asymmetric encryption algorithm named after its inventors. It relies on the mathematical complexity of prime numbers to generate key pairs. It uses a public-private key pair for encryption and decryption, making it suitable for secure data transmission and digital signatures.
- **Elliptic Curve Cryptography (ECC)**
ECC is an asymmetric encryption method based on the mathematical properties of elliptic curves over finite fields. It offers robust security with shorter key lengths than other algorithms, which results in faster computations and lower power consumption.

Common symmetric encryption algorithms :

- **Data Encryption Standard (DES) and Triple DES (3DES)**
IBM® first introduced DES in the 1970s as the standard encryption algorithm, a role it held for many years. However, its relatively short key length (56 bits) made it vulnerable to brute-force attacks, where threat actors try different keys until one works. Triple DES, developed as an enhancement, applies the DES algorithm three times to each data block, significantly increasing the key size and overall security. Eventually, more secure symmetric algorithms replaced both DES and Triple DES.

- **Advanced Encryption Standard (AES)**

AES is generally considered the gold standard of symmetric encryption algorithms. It is widely adopted by organizations and governments worldwide, including the US government. AES offers strong security with key lengths of 128, 192 or 256 bits. Longer key lengths are more resistant to cracking. Specifically, AES-256, which uses a 256-bit key, is known for its high level of security and is often used in highly sensitive situations. AES is also highly efficient in both software and hardware implementations, making it suitable for a wide range of applications.

Common Key Exchange algorithms :

- **Diffie-Hellman**

The Diffie-Hellman key-exchange algorithm is a secure algorithm that offers high performance, allowing two computers to publicly exchange a shared value without using data encryption. This exchanged information is protected with a hash function. The key itself is never exchanged by the two communicating machines, but each machine can generate the identical shared key.

- **RSA for Key Exchange**

Rather than encrypting actual messages (which is slow for large data), RSA is more commonly used to securely transmit a symmetric key (like an AES key), which will then be used for bulk encryption. RSA is also used in digital signatures for authentication and integrity.

Common hash algorithms :

- **MD (Message Digest) Family**

MD2, MD4, and MD5 These methods use a *message digest* (the hash value) that is 128 bits in length. They were created by Ron Rivest and are popularly used for digital signatures.

- **SHA (Secure Hash Algorithm) Family**

Secure Hash Algorithm (SHA) There are several variations on this algorithm, including SHA1, SHA256, SHA384, and SHA512. The differences between them lie in the length of the hash value. SHA was created by a cooperative effort of two U.S. government agencies, NIST and the National Security Association (NSA).

2.1 General introduction to Quantum cryptography

Quantum cryptography is the science of exploiting quantum mechanical properties to perform cryptographic tasks. The advantage of quantum cryptography lies in the fact that it allows the completion of various cryptographic tasks that are proven or conjectured to be impossible using only classical (i.e. non-quantum) communication.

Traditionally, cryptographic security relied on mathematics and took into account the limited computation powers that we have developed. Breaking a cryptographic code would involve factoring extremely large numbers into two primes, typically of over 100 digits in length, which was assumed to be impossible in a reasonable amount of time (less than a million years) even if all of the available computers today worked exclusively on one such problem. However, just because there is no algorithm efficient enough to do the factoring quickly, does not mean that such an algorithm could not be found eventually. Quantum cryptography, which uses photons and relies on the laws of quantum physics instead of "extremely large numbers," is the cutting edge discovery which seems to guarantee privacy even when assuming eavesdroppers with unlimited computing powers. For example, it is impossible to copy data encoded in a quantum state. If one attempts to read the encoded data, the quantum state will be changed due to wave function collapse (no-cloning theorem). This could be used to detect eavesdropping in quantum key distribution (QKD).

2.2 Quantum key distribution (QKD)

Originally theorized in 1984 by Charles H. Bennett (of IBM's Thomas J. Watson Research Center) and Gilles Brassard, quantum key distribution (QKD) is the most common type of quantum cryptography. QKD systems are not typically used to encrypt secure data itself, but rather to make a secure key exchange between two parties by collaboratively building a shared private key, which can in turn be used for traditional symmetric key encryption methods.

QKD systems work by sending individual photon light particles across a fiber optic cable. This stream of photons travels in a single direction and each one represents a single bit, or qubit, of data—either zero or one. Polarized filters on the sender's side change the physical orientation of each single photon to a specific position. And the receiver uses two available beam splitters to read the position of each photon as they're received. The sender and receiver compare the sent photon positions to the decoded positions, and the set that matches becomes the key.

To better understand QKD, imagine two people, Alice and Bob, who need to establish a secure connection. They can use QKD to create a secure cryptographic key by sending polarized photons over a fiber optic cable. The cable doesn't need to be secured because each photon will have its own randomized quantum state. Should someone, let's call her Eve, be eavesdropping, Alice and Bob will always be able to tell because it's impossible to observe a quantum state without also affecting it. In this way, QKD systems are considered to be unhackable. If Bob and Alice detect a change in the quantum states of the photons, they'll know that Eve is eavesdropping. And if Eve is eavesdropping, Bob and Alice will always be able to detect it.

Although the benefits of QKD have been proven in both laboratory and field settings, there are many practical challenges preventing widespread adoption, most notably infrastructure requirements. Photons sent across fiber optic cables degrade over distances of about 248 to 310 miles. However, recent advancements have extended the range of some QKD systems across continents by using secure nodes and photon repeaters.

2.3 E91 key distribution protocol

The E91 protocol, also known as the Ekert 91 protocol, is a quantum cryptography protocol proposed by physicist Artur Ekert in 1991. It is designed to enable secure communication between two parties, typically referred to as Alice and Bob, by utilizing the phenomenon of quantum entanglement.

2.3.1 Why E91

Over the years many key distribution protocols have been developed , One of the most well-known is the pioneering BB84 . While groundbreaking , it has a notable flaw , it needs a secure channel where Alice sends her qubits to Bob . An eavesdropper as Eve , can intercept the channel and capture the qubit Alice send , although due to the no cloning theorem , quantum states cannot be copied , thus she cant just replicate it and send it to Bob . What she can do though is measure it with the base she has chosen and send to Bob her result. . Another flaw of BB84 is the lack of eavesdrop detection unlike E91 which provides a robust detection mechanism.

In BB84 , if the 2 keys we produced don't match up , we discard them , and we suspect noise disturbance or eavesdrop , which isn't the most effective solution . In a scenario where Eve was unlucky and she measured the qubit in the wrong base , while Alice and Bob had different bases , Alice and Bob will drop it due to mismatch basis . In that case the mistake that eve made went unnoticed , which lead Alice and Bob to not suspect eavesdrop .

Our point here is when Alice and Bob have different bases , Eve goes untraceable since their measurements are discarded . In order to trace eavesdrop , Alice and Bob must have the same base and Eve must measure in the wrong base.

In that case they will include their measurements , and by comparing their final keys , they will see that they don't match , suspecting noise disturbance or eavesdrop . Either way if the keys have more different bits than the tolerance they will discard them and start over again .

E91, based on the properties of entanglement, provides a more robust method of key distribution that does not require a secure channel and makes it impossible for an eavesdropper to go undetected as we will see later.

2.3.2 How E91 works

In order to examine E91 we will divide it into two sub processes, the key creation, and the entanglement verification.

2.3.2.A Key creation

The first thing Alice and Bob have to do is establish a key. They start by sharing an entangled state distributing multiple copies of it between them.

We suppose that Alice and Bob share the following entangled state:

$$|\psi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)_{AB}$$

Due to $|\psi^+\rangle$, if Alice and Bob measure in the same basis then the outcome will be perfectly anti-correlated if measured in Z or perfectly correlated if measured in X, with each outcome occurring with equal (uniform) probability.

Let's prove it, first for X:

$$\begin{aligned} |\psi^+\rangle_{AB} &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)_{AB} = \frac{1}{\sqrt{2}} \left[\frac{1}{2}(|+\rangle + |-\rangle) \otimes (|+\rangle - |-\rangle) + \frac{1}{2}(|+\rangle - |-\rangle) \otimes (|+\rangle + |-\rangle) \right] \\ &= \frac{1}{2\sqrt{2}}(2|++\rangle - 2|--\rangle) = \frac{1}{\sqrt{2}}|++\rangle - \frac{1}{\sqrt{2}}|--\rangle \end{aligned}$$

So we see that:

$$\text{Prob}(|++\rangle_{AB}) = \frac{1}{2}, \quad \text{Prob}(|--\rangle_{AB}) = \frac{1}{2}$$

$$\text{Prob}(|+-\rangle_{AB}) = 0, \quad \text{Prob}(|-+\rangle_{AB}) = 0$$

Therefore if we map $|+\rangle$ to 0 and $|-\rangle$ to 1 then we can see correlation

If Alice measures a 0 Bob will always measure a 0

If Alice measures a 1 Bob will always measure a 1

Now let's prove it for Z:

$$\text{Prob}(|01\rangle_{AB}) = \frac{1}{2}, \quad \text{Prob}(|10\rangle_{AB}) = \frac{1}{2}$$

$$\text{Prob}(|00\rangle_{AB}) = 0, \quad \text{Prob}(|11\rangle_{AB}) = 0$$

If Alice measures a 0 Bob will always measure a 1

If Alice measures a 1 Bob will always measure a 0

Due to anticorrelation unlike previously that the results were correlated, Bob will have to use a X operator to find the exact outcome as Alice, in order to flip it.

By repeating this process many times we make each bit of the key.

2.3.2.B Entanglement verification

As we said in the beginning E91 offers a robust solution in detecting if eavesdrop happened . The entangled pair serves two purposes based on Alice's and Bob's choices of base. When they have the same ones , as we analyzed it in 2.3.2.A , it serves creation purposes . When they have different ones which is the verification case , instead of dropping those qubits as in BB84 , we utilize the fundamental of monogamy of entanglement and use them as a checker to test the security of our system .

What is the monogamy of entanglement

Monogamy of entanglement is a fundamental property of quantum states that constraints how correlated multiple qubits can be. So in particular to our example if Alice and Bob know that they share a maximally entangled state then they cannot be correlated with eve at all, meaning that eve cannot use their key. So we can also say that the less the entanglement between Alice and Bob ,the more correlated they are with the eavesdropper and vice versa .

Now lets see how is Alice and Bob able to verify the entanglement of their shared pair. In order to do that we have to use the CHSH inequality

CHSH inequality

To understand it first we will analyze it in classical version and then in the quantum .

Classical Version

We start by considering 4 classical variables: A,A',B,B' each taking values ± 1 . Then, the expression:

$$C = A(B + B') + A'(B - B')$$

For every possible value combination , we have

$$C = \pm 2$$

Now, suppose we repeat this process n times and define the average value:

$$S = \langle A(B + B') \rangle + \langle A'(B - B') \rangle$$

This leads directly to the **CHSH inequality** in terms of correlation functions:

$$S = \langle AB \rangle + \langle AB' \rangle + \langle A'B \rangle - \langle A'B' \rangle \leq 2$$

This form **must hold** if the correlations are explained by local hidden variable theory..

Quantum Version

But how does the CHSH inequality applies on quantum mechanics. Well lets consider Alice and Bob sharing a $|\psi\rangle$ and we have some measure observables A, A', B, B' , where A, A' are Alice's measurement basis on the outcome of the state $|\psi\rangle$ and likewise for Bob.

The expectation value of the product of measurements becomes:

$$\langle AB \rangle = \langle \psi | A \otimes B | \psi \rangle$$

$$S = |\langle AB \rangle + \langle AB' \rangle + \langle A'B \rangle - \langle A'B' \rangle| \leq 2$$

If $|S| \leq 2$ there either has been interception from Eve or a mess up from our equipment so the key is dropped we start again. If the key is approximately $2\sqrt{2}$ there is no interception the photons are entangled and the key is kept.

This method is exemplary strong because it offers precise and mathematical means to counter eavesdropping or experimental problems without any disclosure of the cryptographic key .

What is specifically remarkable is that, based on classical assumptions, we through the CHSH inequality have proven that no classical random variables could yield the result $|S| > 2$. This gives way in which quantum mechanics allows for some states, namely, highly entangled states, to violate this classical boundary.

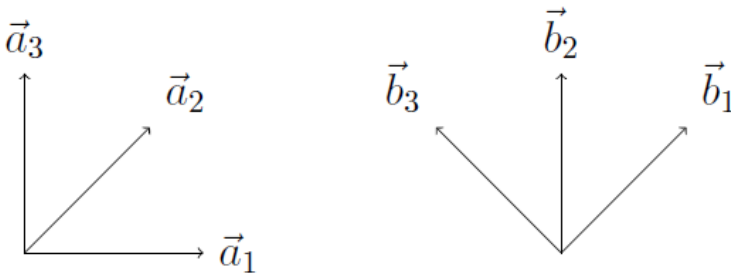
So by checking if the state violates the CHSH inequality, Alice and Bob will know if their shared state is maximally entangled .



Alice and Bob use 3 measurement basis each, because we need same basis to create a key and also different ones to find if the shared state is maximally entangled and detect eavesdrop. Most common is two same and one different, a ratio 2/1 .

2.3.3 Example

Alice and bob measure the photons in their respective bases $\{a_1=0^\circ, a_2=45^\circ, a_3=90^\circ\}$, $\{b_1=45^\circ, b_2=90^\circ, b_3=135^\circ\}$. We see 2 matching pairs $a_2 = b_1$ and $a_3 = b_2$ and one unmatched pair a_1, b_3 . We visualize it as below.



After they receive multiple copies of the shared state they apply their randomly chosen measurement basis on their outcomes

Alice basis	A2	A3	A2	A2	A1	A3
Bob basis	B2	B3	B1	B2	B1	B2

In order to detect eavesdropping Alice and Bob use the remaining measured photons 2/3 to check the CHSH inequality .

$$S = |\langle AB \rangle + \langle AB' \rangle + \langle A'B \rangle - \langle A'B' \rangle|$$

If $|S| \leq 2$ there either has been interception from Eve or a mess up from our equipment so the key is dropped we start again. If the key is approximately $2\sqrt{2}$ there is no interception the photons are entangled and the key is kept.

Qiskit experiment :

Setup:

Our source will provide us with the above bell state :

$$|\psi_s\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B - |1\rangle_A \otimes |0\rangle_B) = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

For our inconvenience we will use the same basis from above . There ideal because , due to their non linearity and their 45 degree difference between them , we will achieve maximum CHSH violation .

$$\begin{aligned} b_i = 1 : \quad \vec{a}_1 &= (1, 0, 0) \quad (X \text{ observable}) & b'_j = 1 : \quad \vec{b}_1 &= \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) \quad (W \text{ observable}) \\ b_i = 2 : \quad \vec{a}_2 &= \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) \quad (W \text{ observable}) & b'_j = 2 : \quad \vec{b}_2 &= (0, 0, 1) \quad (Z \text{ observable}) \\ b_i = 3 : \quad \vec{a}_3 &= (0, 0, 1) \quad (Z \text{ observable}) & b'_j = 3 : \quad \vec{b}_3 &= \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) \quad (V \text{ observable}) \end{aligned}$$

Then we can rewrite our basis w,v in the form of X,Z :

$$\begin{aligned} W &= \frac{1}{\sqrt{2}}(X + Z), \quad V = \frac{1}{\sqrt{2}}(-X + Z) \\ \langle X \otimes W \rangle_{\psi_s} &= -\frac{1}{\sqrt{2}}, \quad \langle X \otimes V \rangle_{\psi_s} = \frac{1}{\sqrt{2}}, \\ \langle Z \otimes W \rangle_{\psi_s} &= -\frac{1}{\sqrt{2}}, \quad \langle Z \otimes V \rangle_{\psi_s} = -\frac{1}{\sqrt{2}}. \end{aligned}$$

So now lets calculate the CHSH :

$$C = \langle X \otimes W \rangle - \langle X \otimes V \rangle + \langle Z \otimes W \rangle + \langle Z \otimes V \rangle = -2\sqrt{2}.$$

We have maximum entanglement , due to the choice of basis we make above and the fact we use an epair .

Then Alice and Bob will generate random strings of basis to do their measurements :

$$b = (b_1, \dots, b_N), \quad b' = (b'_1, \dots, b'_N), \quad \text{where } b_i, b'_j \in \{1, 2, 3\}.$$

An eavesdropper Eve, who attempts to intercept makes her random string basis , from the available basis of each party

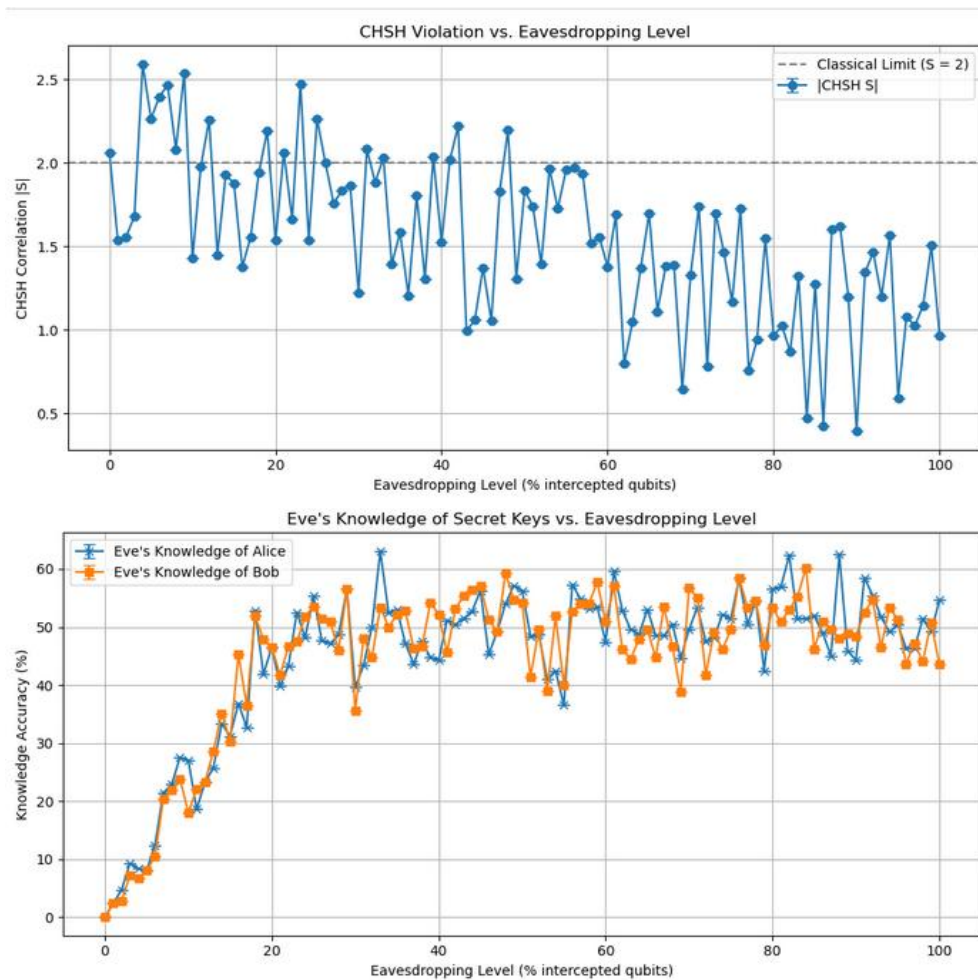
$$e = [(e_1, e'_1), (e_2, e'_2), \dots, (e_M, e'_M)]$$

The bit strings will be used later on , when we will create our test circuits , to simulate the protocol and make our measurements.

In order to simulate an variable eavesdropping rate , eve will not take place In every round in each key creation attempt . For example if we aim to make 500 bit key with 10% rate of eavesdropping se will take place in only 50 of them attempts .

We do so to monitor how drastically her actions effect the chsh inequality and see when is detectable , if so .

We repeat the process multiple times, recording each party's measurement outcomes , the result of the CHSH inequality and the information she may have gained about Alice's and Bob's key.

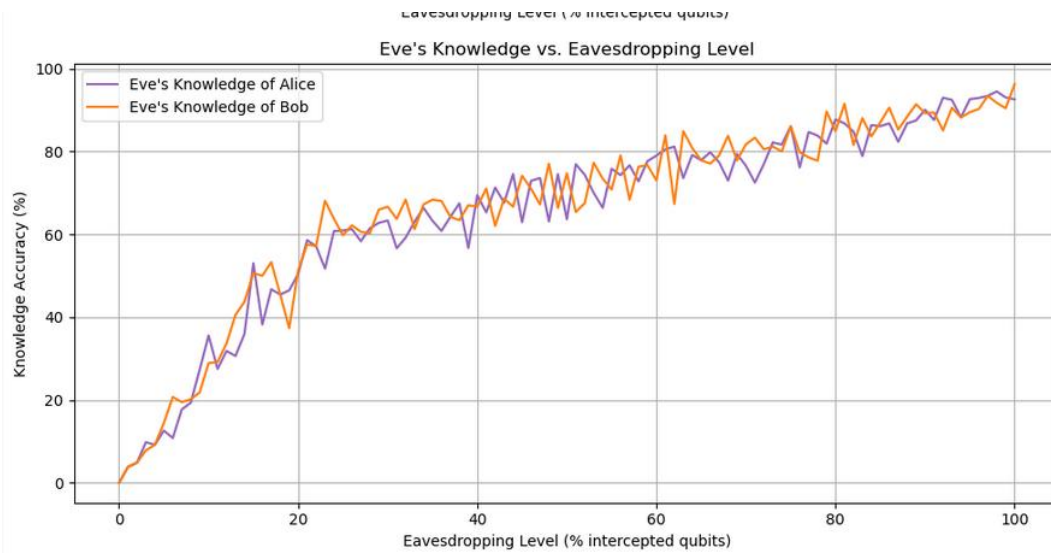
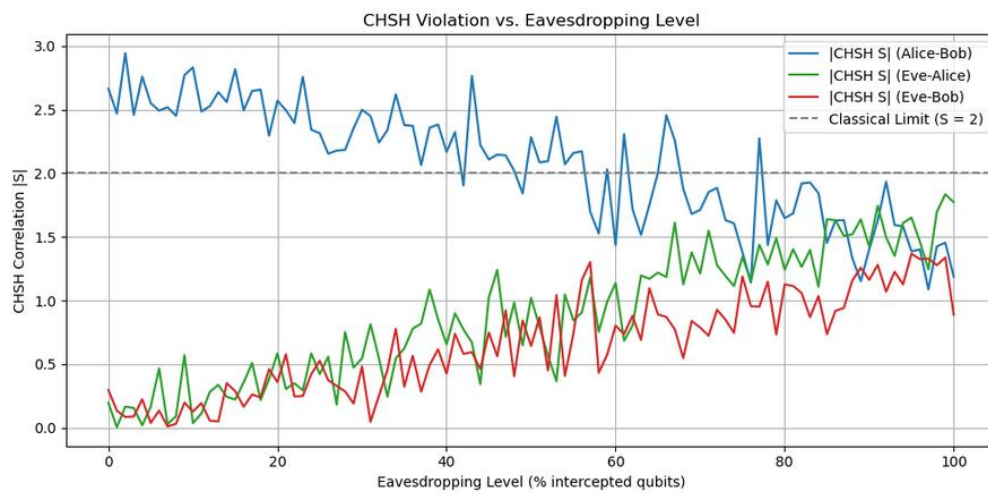


Before we let eve choose independently for either Alice's or Bob's Base (from their entire collection) . This meant that she can choose different base for each one . Now we will give her an advantage , we will gave her the two sets that Alice and Bob have in common and she will be able to choose one of each , in each repeat (Eve chooses between **W** and **Z** basis for both) . In that way when Alice and Bob are matching she has 50% chance she matches them .

Further more in this experiment except the correlation between Alice and Bob , we will monitor the correlation they have with Eve .

$$S_{\text{Eve-Alice}} = E(W, X) + E(W, W) + E(Z, X) - E(Z, W)$$

$$S_{\text{Eve-Bob}} = E(W, V) + E(W, Z) + E(Z, V) - E(Z, Z)$$



What we learn from our experiment :

When Eve knows the collection of basis of each party and chooses independent and random for each one :

- Even at a **very low eavesdropping rate**, the correlation between **Alice and Bob** drops significantly, making her presence easy to spot through CHSH violations.
- At **full interception**, Eve still fails to gather enough information to accurately reconstruct the shared keys. With only about **60% knowledge** and no information about which bits are actually valid **brute-forcing the key becomes impractical**. That does not mean Alice and Bob should keep it .

When Eve uses only the matching sets :

- Catching Eve gets more challenging , since she manage to reduce the rate at which the CHSH correlation drops, making her presence less obvious. However, even with an improved approach, Eve can never reach the quantum maximum $2\sqrt{2}$, since she collapses the epair .
- Another noticeable is that the less correlated Alice and Bob get, the more they get with Eve but still they can never reach $2\sqrt{2}$ since they don't share an epair , and they don't have 45 or 22.5 degree difference in the basis the use for the chsh..

Conclusion

As established at the outset, the security of a cryptographic system fundamentally depends on the secrecy of the key. Consequently, it is imperative to invest in robust key distribution mechanisms. Currently, cryptographic security relies on two primary factors: the generation of large pseudo-random numbers and the computational infeasibility of breaking the encryption within a reasonable time frame using existing technology.

Quantum Key Distribution protocols such as E91, might be a robust solution with its ability of generating truly random keys and ensuring secure key exchange. However, despite its theoretical strengths, QKD is very vulnerable in denial-of-service attacks. Eve can repeatedly attempt to eavesdrop on the quantum channel. Due to detection of eavesdropping, Alice and Bob are forced to drop their keys. If Eve persistently eavesdrops, Alice and Bob will be unable to establish a shared secret key, effectively rendering the system unusable.

This form of attack, while not compromising the secrecy of the key itself, poses a critical threat to the availability of secure communication. Given the historical impact of DoS attacks, which have demonstrated the potential for widespread disruption, it is evident that similar vulnerabilities in QKD protocols must be addressed. The design of quantum cryptographic systems should embed a mechanism to handle such attacks.

Sources

https://link.springer.com/referenceworkentry/10.1007/978-1-4419-5906-5_487
<https://www.ibm.com/think/topics/asymmetric-encryption>
<https://www.ibm.com/think/topics/symmetric-encryption>
<https://www.sciencedirect.com/topics/computer-science/hashing-algorithm>
https://www.eclass.tuc.gr/modules/document/file.php/HMMY304/Books/ebooksclub.org__Quantum_Computing_Explained.pdf
<https://cs.stanford.edu/people/eroberts/courses/soco/projects/2004-05/cryptography/quantum.html>
https://en.wikipedia.org/wiki/Quantum_cryptography
<https://arxiv.org/pdf/2409.16284>
<https://eprint.iacr.org/2020/1074.pdf>
<https://arxiv.org/html/2409.07597v1#S3>

Video

<https://www.youtube.com/watch?v=oDz4hDsJQFw>
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<https://youtu.be/V3WzH2up7Os?si=GYCFoMxlXF2h8UWO>

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